

Tracking Evolving Exploit Discourse in Hacker Communities: A Diachronic Exploit Text Graph Framework for Proactive Cyber Threat Intelligence

*Cybersecurity threats emerge and evolve in hacker communities long before they are formally catalogued in vulnerability databases, yet existing threat-intelligence systems process static snapshots of hacker discourse, missing the temporal semantic drift that signals emerging attack patterns. We present the **Exploit Text Graph (ETG)**—a cumulative temporal co-occurrence graph of exploit vocabulary—and the **Diachronic Graph Transformer (DGT)**, a masked graph-attention model that embeds ETG nodes across successive temporal spells and predicts which terms will undergo the greatest semantic shift in the next observation window. Applied to HackerSignal, a 1,783,276-post multi-source corpus spanning 12 temporal spells from January 2016 through April 2026, DGT achieves a heldout mean absolute error of 0.0774 on next-spell shift prediction, a 24.5% reduction over a naive mean baseline and significantly better than 20 of 38 benchmark models—including GloVe, PPMI, SGNS, and five dynamic graph neural network baselines—at a false-discovery-rate-corrected $q < 0.05$. Ablation experiments validate four design principles: cumulative graph encoding, masked temporal attention, a shift-predictive training objective, and multi-source corpus construction. A six-term case study traces semantically pivotal cybersecurity events—from the Mirai botnet and WannaCry through the emergence of Ransomware-as-a-Service and LLM jailbreaking—demonstrating that DGT detects vocabulary*

realignment in real time and produces an actionable forward-looking watchlist of predicted next-spell shifts. Theoretical contributions, design principles, and practitioner implications for proactive cyber threat intelligence are discussed.

Keywords: cyber threat intelligence, design science research, diachronic graph transformer, exploit text graph, hacker forum analytics, temporal semantic shift

Tracking Evolving Exploit Discourse in Hacker Communities: A Diachronic Exploit Text Graph Framework for Proactive Cyber Threat Intelligence

1 INTRODUCTION

The cybersecurity threat landscape is defined by an asymmetric detection lag: adversaries develop, refine, and share novel attack techniques in hacker forums long before those techniques enter the formal record of known vulnerabilities. By the time a vulnerability receives a Common Vulnerabilities and Exposures (CVE) identifier and appears in catalogues such as the CISA Known Exploited Vulnerabilities (KEV) list, security teams face a compressed remediation window—often measured in days (CISA, 2024; Jacobs et al., 2021). Our analysis of 1,783,276 hacker-forum posts collected between January 2016 and April 2026 reveals that 271,275 distinct CVE identifiers were mentioned in hacker discourse, reflecting a vast pool of vulnerability intelligence that circulates in underground communities well before, during, and after formal classification. Bridging this information gap is a central challenge for proactive cyber threat intelligence (CTI).

Hacker communities are early-warning systems for emerging threats. Prior research in IS cybersecurity analytics has demonstrated that dark-web forums, hacker boards, and underground marketplaces contain signals predictive of real-world exploit activity (Ebrahimi et al., 2022; Nunes et al., 2016; Samtani et al., 2017, 2022). These communities develop, contest, and evolve a shared technical vocabulary—a lexicon in which terms not only denote artifacts (tools, vulnerabilities, techniques) but also encode the community’s collective understanding of how those artifacts relate to one another. The vocabulary itself is dynamic: the word *jailbreak* referred exclusively to iOS

firmware bypass through 2022 before migrating, almost entirely, to large-language-model (LLM) safety-filter circumvention following the public release of ChatGPT. The word *stealer* shifted from a generic adjective to a recognised product category (e.g., RedLine Stealer, Raccoon Stealer) with versioned releases and subscription pricing. Tracking these *semantic shifts*—changes in how terms are used relative to their co-occurrence neighbours—offers a mechanism for detecting emerging attack paradigms before they crystallise into named CVEs.

Despite this potential, existing IS cybersecurity analytics systems treat hacker discourse as a static or bag-of-words phenomenon. Text-mining methods (Samtani et al., 2017; Tavabi et al., 2018), deep transfer learning (Samtani et al., 2022), and cross-lingual models (Ebrahimi et al., 2022) extract threat signals from individual snapshots or aggregate corpora, but none model the continuous temporal drift of exploit vocabulary as a predictive signal. Classical diachronic semantic models (Hamilton et al., 2016; Kim et al., 2014) capture word-level drift but discard the relational co-occurrence structure that encodes how exploit techniques combine—a structure central to understanding attack chains. Dynamic graph neural networks (Rossi et al., 2020; Sankar et al., 2020) model temporal graph evolution but are not designed for the shift-prediction task or for the cumulative nature of exploit knowledge. The result is a methodological gap: no prior IS artifact simultaneously (1) represents exploit discourse as a relational graph, (2) learns diachronic embeddings across sequential graph snapshots, and (3) forecasts next-period semantic shift to support proactive CTI.

We address this gap through a Design Science Research (Hevner et al., 2004; Peffers et al., 2007) programme that produces two interlocking IS artifacts. The **Exploit Text Graph (ETG)** is a cumulative directed weighted graph in which nodes represent stemmed exploit terms and edges represent co-occurrence within a five-token sliding window across all posts up to and including

each temporal spell. The ETG grows monotonically—capturing the accumulation of community knowledge—and is constructed at 12 spell boundaries spanning a decade of hacker discourse. The **Diachronic Graph Transformer (DGT)** is a masked graph-attention encoder trained on the sequence of ETG snapshots to produce per-term embeddings whose cosine distance between consecutive spells quantifies semantic shift. A lightweight prediction head learns to forecast this shift for the forthcoming spell, enabling the construction of a forward-looking watchlist of terms expected to undergo the greatest semantic realignment.

This paper makes three primary contributions to IS research and practice. First, we contribute two novel IS artifacts—the ETG and DGT—accompanied by four empirically validated design principles. Second, we contribute a rigorous 38-model benchmark evaluation spanning non-neural diachronic baselines, static and dynamic graph embeddings, and modern graph transformers, with paired significance testing and false-discovery-rate correction. Third, we contribute a six-term case study grounded in publicly verifiable cybersecurity events, demonstrating that DGT correctly identifies semantic pivots associated with the Mirai botnet, WannaCry/EternalBlue, Ransomware-as-a-Service emergence, info-stealer marketplaces, Ivanti zero-days, and LLM jailbreaking—each a shift that a practitioner would recognise as a meaningful change in the threat landscape.

The remainder of this paper is organised as follows. Section 2 reviews related work across four theoretical streams. Section 3 describes the Design Science Research approach. Section 4 presents the artifact design. Section 5 reports the benchmark evaluation. Section 6 presents the case study. Section 7 discusses theoretical contributions, practical implications, and limitations. Section 8 concludes.

2 THEORETICAL BACKGROUND AND RELATED WORK

2.1 IS Cybersecurity Analytics and Hacker Community Intelligence

Hacker communities—including dark-web forums, surface-web hacker boards, and underground marketplaces—have become a primary data source for proactive cybersecurity analytics in the IS literature. Foundational work by Samtani et al. (2017) introduced a hacker asset framework that identifies emerging tools, exploits, and key actors in hacker forums, demonstrating that community-level text analytics can surface intelligence unavailable from traditional vulnerability feeds. Subsequent work extended this framework to link dark-web exploit postings to known CVEs using semantic embedding techniques, showing that deep representations of hacker posts can predict exploitation activity (Samtani et al., 2022). Cross-lingual extensions (Ebrahimi et al., 2022) addressed the multilingual character of global hacker communities, applying zero-shot and adversarial learning methods to non-English forums, while vulnerability prioritisation research (Ullman & Samtani, 2024) demonstrated that multi-view learning over vulnerability metadata improves CVSS scoring and KEV classification.

A parallel literature focuses on SCADA and IoT vulnerability intelligence, extracting structured threat indicators from unstructured hacker posts (Samtani et al., 2020), and on the epidemiology of exploit diffusion through underground networks (Nunes et al., 2016; Tavabi et al., 2018). Collectively, this body of work establishes that hacker communities are rich, timely, and idiosyncratic sources of threat intelligence—but it also reveals a shared limitation: all existing IS cybersecurity analytics methods process hacker discourse at a single point in time or aggregate across time, treating the corpus as a static collection. None model how the *meaning* of exploit terms evolves continuously, nor do they forecast which terms are approaching a semantic transition. Our

work directly addresses this gap.

2.2 Diachronic Semantic Models

Diachronic (time-varying) semantic models seek to track how the meaning of words changes across corpora collected at different time periods. The foundational approach of Hamilton et al. (2016) trains Word2Vec embeddings (Mikolov et al., 2013) on successive temporal corpora and aligns them using Procrustes rotation, yielding cosine distances that measure per-word semantic change. Kim et al. (2014) extended this approach with temporal neural language models that explicitly condition word representations on time. Kulkarni et al. (2015) proposed statistical tests for detecting significant linguistic change in large corpora. More recently, Röttger and Pierrehumbert (2021) demonstrated that fine-tuning BERT (Devlin et al., 2019) on time-stamped data improves temporal adaptation on downstream tasks, though at the cost of retraining for each new time period.

These methods share a fundamental limitation for the CTI application: they operate at the word level and discard the relational structure of co-occurrence. In hacker discourse, meaning is conveyed not only by individual terms but by the combinations of terms that co-occur—*ransomware* and *affiliate* co-occurring with *panel* signals a different context than *ransomware* co-occurring with *decryption*. A graph representation captures this combinatorial semantics explicitly, enabling the model to detect when a term’s neighbourhood changes—which is, precisely, what semantic shift means in a community of practice. Furthermore, diachronic word embedding methods are designed to measure shift, not to *predict* it. Our DGT introduces a shift-prediction head that forecasts next-spell cosine distances, enabling proactive rather than retrospective analysis.

2.3 Graph Neural Networks and Dynamic Graph Learning

Graph neural networks (GNNs) generalise deep learning to graph-structured data by propagating information across node neighbourhoods (Hamilton et al., 2017; Kipf & Welling, 2017; Veličković et al., 2018). Graph embedding methods—DeepWalk (Perozzi et al., 2014), Node2Vec (Grover & Leskovec, 2016), LINE (Tang et al., 2015)—learn static node representations by factorising adjacency or applying random walks, providing strong baselines for co-occurrence graph representation. For temporal graphs, dynamic graph neural networks model evolving node states: DySAT (Sankar et al., 2020) applies self-attention across structural and temporal dimensions; EvolveGCN (Pareja et al., 2020) uses recurrent units to evolve GCN weight matrices; TGN (Rossi et al., 2020) combines memory modules with graph attention; and JODIE (Kumar et al., 2019) projects user and item trajectories through time. These models have been applied to link prediction and node classification on interaction graphs, but not to the diachronic co-occurrence graph setting or the shift-prediction objective.

Graph Transformers adapt the self-attention mechanism (Vaswani et al., 2017) to graph inputs. GraphGPS (Rampášek et al., 2022) integrates local MPNN layers with global attention and Laplacian positional encodings (Kreuzer et al., 2021). TokenGT (Kim et al., 2022) treats graphs as sequences of node and edge tokens, enabling full quadratic attention. These architectures excel on molecular property prediction and graph classification benchmarks, but have not been applied to temporal co-occurrence graphs in the NLP or security domain. Our DGT adapts the graph transformer paradigm specifically for diachronic exploit graph embedding by introducing masked temporal attention—a causal constraint that prevents future-spell information from leaking into past-spell representations—and a shift-predictive training objective absent from all prior graph

transformer formulations.

2.4 The ETG and DGT Paradigm: Filling the Gap

Table 1: Comparison of Related Work on Key Dimensions					
Approach	Representative Works	IS Domain	Graph	Temporal	Shift Pred.
IS CTI Analytics	Samtani et al. (2017, 2022)	✓	×	×	×
Diachronic Semantics	Hamilton et al. (2016); Kim et al. (2014)	×	×	✓	×
Dynamic GNNs	Rossi et al. (2020); Sankar et al. (2020)	×	✓	✓	×
Graph Transformers	Rampášek et al. (2022); Kim et al. (2022)	×	✓	×	×
ETG + DGT (ours)	This paper	✓	✓	✓	✓
<i>Note: ✓ = supported; × = not supported; ~ = partially supported. IS Domain refers to hacker-community or cybersecurity analytics; Graph Structure to relational co-occurrence representation; Temporal to sequential spell modelling; Shift Prediction to forecasting next-period semantic change.</i>					

As shown in Table 1, no prior work occupies all four dimensions simultaneously. The ETG fills the relational IS-domain gap by constructing a cumulative graph of exploit co-occurrences grounded in hacker forum data. The DGT fills the temporal shift-prediction gap by applying masked graph-attention across ETG snapshots and training explicitly to forecast next-period cosine drift. Together, they instantiate four design principles derived from this gap analysis:

- **DP1 — Cumulative Encoding:** Exploit knowledge accumulates; each spell’s graph should encode all prior co-occurrence evidence, not only the current period.
- **DP2 — Masked Temporal Attention:** Future spell embeddings must not leak into past representations; temporal attention must be causally masked.
- **DP3 — Shift-Predictive Objective:** The model must be trained to forecast next-spell cosine shift directly, not to reconstruct the graph or classify nodes.

- **DP4 — Multi-Source Corpus:** Threat intelligence requires breadth; the corpus must combine hacker communities, vulnerability references, advisories, exploit archives, and related sources.

3 DESIGN SCIENCE RESEARCH APPROACH

3.1 DSR Methodology

We adopt the Design Science Research methodology of Hevner et al. (2004), which frames IS research as the creation and evaluation of purposeful artifacts that address organisational problems. The three DSR cycles—relevance, design, and rigor—structure our work. The relevance cycle grounds the research in the practitioner need for proactive CTI. The design cycle encompasses the construction and instantiation of the ETG and DGT artifacts. The rigor cycle connects the artifacts to the prior knowledge base in diachronic semantics, graph learning, and IS cybersecurity analytics. We additionally apply the six-step DSRM process of Peffers et al. (2007): problem identification, objectives of a solution, design and development, demonstration, evaluation, and communication (Gregor & Hevner, 2013).

3.2 Problem Relevance

The practitioner problem is well-documented: CTI analysts at security operations centres, CISOs at enterprise organisations, and analysts at Information Sharing and Analysis Centres (ISACs) require timely intelligence about emerging threats. Existing tools—SIEM platforms, signature-based intrusion detection, static vulnerability feeds—are reactive; they alert after an attack technique is known and catalogued. Hacker forums contain this information earlier, but no IS artifact operationalises the temporal dimension of that intelligence. The problem is an IS problem—not

a purely technical one—because it involves the design of an artifact for a knowledge-work context (CTI analysis), requires organisational data (multi-source hacker corpora), and has implications for managerial decision-making (vulnerability prioritisation, resource allocation for patching).

3.3 Design Objectives

Three design objectives flow directly from the problem and the gap analysis in Section 2.4. First, the artifact must *represent exploit relationships* as a graph, capturing how terms co-occur and combine. Second, it must *model temporal drift* across successive observation periods, learning embeddings that encode semantic evolution. Third, it must *generate actionable predictions*—a ranked watchlist of terms predicted to shift meaning in the next period—that a CTI analyst can operationalise without additional modelling expertise. These objectives map directly to the ETG, the DGT encoder, and the watchlist generation procedure, respectively.

3.4 Evaluation Strategy

Following Hevner et al. (2004)’s taxonomy of evaluation modes, we employ two complementary approaches. First, a *technical evaluation* benchmarks DGT against 38 models across four experiment tiers—non-neural diachronic baselines, graph embedding and dynamic GNN baselines, modern graph transformers, and DGT component ablations—with paired significance testing and Benjamini–Hochberg false discovery rate correction (Benjamini & Hochberg, 1995). This evaluation addresses the question of whether DGT advances the state of the art on the shift-prediction task. Second, a *naturalistic evaluation* presents a six-term case study anchored to publicly verifiable real-world cybersecurity events, demonstrating the artifact’s practical utility without requiring a controlled laboratory study. A formal user study with CTI practitioners is

identified as priority future work.

4 ARTIFACT DESIGN

4.1 HackerSignal Corpus

4.1.1 The Full HackerSignal Dataset

The ETG analysis draws on HackerSignal (Ampel & Samtani, 2026), a large-scale multi-source benchmark corpus linking hacker community discourse to the CVE vulnerability lifecycle. The full HackerSignal release comprises **7,447,646 exact-deduplicated documents** from **64 public forum and source identifiers** spanning **eight source layers** and a **36-year window** (1991–2026):

1. **Hacker Community (6,966,014 docs; 56 forums):** Organic security discourse from darkweb marketplaces, security discussion boards, and multilingual hacker communities. Major forums include `gayanku_darkweb` (3.11M rows, 2011–17), `antichat` (1.82M, 2002–26), `evolution_darkweb` (471K, 2014–15), `antionline` (463K, 2001–26), `persiantools` (280K), `wwhclub` (161K), and `hackforums` (151K, 2009–15), alongside 48 additional forums.
2. **Vulnerability Reference (340,536 docs):** National Vulnerability Database (NVD) CVE descriptions (2002–26), each with a CVE identifier; the retrieval corpus for HackerSignal’s benchmark tasks.
3. **Exploit Archive (46,653 docs):** ExploitDB, 0day.today, PacketStorm, ZeroScience, and Vulnerability-Lab proof-of-concept exploit descriptions (1991–2026).
4. **Exploit Q&A Reference (46,505 docs):** Structured exploit question-and-answer references

(circa 2000).

5. **Advisory Reference (28,684 docs):** GitHub Advisory, CISA advisories, and CERT notices (2017–2026).
6. **Bug-Bounty Disclosure (9,453 docs):** HackerOne and related bug-bounty platform reports (2013–2024).
7. **Fix-Commit Reference (8,293 docs):** CVEfixes security-relevant fix commit messages (1999–2022).
8. **Exploitation Reference (1,508 docs):** CISA Known Exploited Vulnerabilities (KEV) catalog entries (2021–2026).

Documents are unified through a shared CVE identifier namespace and a common JSONL record format carrying source provenance, ISO 8601 timestamps, and pseudonymized author hashes. Across all 7.45M documents, 4.8% carry explicit CVE references, 23.5% contain at least one CTI keyword, and estimated near-duplicates account for 8.2% (addressed through SHA-256 exact deduplication and MinHash-LSH near-deduplication during corpus construction). The corpus spans multilingual content—including Russian, Turkish, Persian, Arabic, and Chinese in DeepDarkCTI and darknet sources—though English-language posts constitute the substantial majority.

4.1.2 ETG Corpus: Subsetting HackerSignal to the Analysis Window

The ETG analysis requires a *temporally coherent* corpus aligned with the modern era of systematic CVE tracking and structured exploit discourse. We apply a single temporal filter: **January 1,**

2016 onward. The 2016 boundary is motivated by three converging factors. First, the CISA KEV catalog, the NVD’s modern structured CVE format, and the major hacker community platforms that dominate our analysis all reach maturity and consistent coverage after 2016. Second, the pre-2016 period is dominated by a small number of historical forum archives (Gayanku 2011–17, Evolution 2014–15) whose temporal coverage is asymmetric with respect to the other source layers. Third, the ETG’s design is oriented toward the *modern* exploit discourse era—the decade in which IoT weaponisation, ransomware-as-a-service, nation-state zero-day exploitation, and LLM safety bypass all emerged as dominant threat paradigms.

Applying the 2016+ filter to HackerSignal yields the **ETG corpus of 1,783,276 documents** across **seven active source layers** (Table 2). The Exploit Q&A Reference layer (46,505 documents, circa 2000) falls entirely outside the analysis window and is excluded. All other layers are retained but reduced in size relative to the full HackerSignal release; the largest reduction occurs in the Hacker Community layer (6,966,014 → 1,455,413), primarily because Gayanku (2011–17), Evolution (2014–15), HackForums (2009–15), and other historical archives largely predate our cutoff.

Table 2: ETG Corpus Source Layer Statistics (HackerSignal Subset, 2016–2026)				
Source Layer	ETG Corpus	%	Full HackerSignal	Primary Content
Hacker Community	1,455,413	81.6	6,966,014	Forum posts, tool releases, community discourse
Vulnerability Reference	270,515	15.2	340,536	NVD CVE descriptions
Advisory Reference	28,684	1.6	28,684	GitHub Advisory, CISA advisories
Exploit Archive	11,652	0.65	46,653	PoC exploit descriptions (ExploitDB, etc.)
Bug Bounty Disclosure	8,338	0.47	9,453	HackerOne reports
Fix Commit Reference	7,166	0.40	8,293	Security fix commit messages
Exploitation Reference	1,508	0.08	1,508	CISA KEV entries
ETG Total	1,783,276	100.0	7,447,646	
<i>Note: The ETG corpus is derived from HackerSignal (Ampel & Samtani, 2026) by applying a January 2016 temporal cutoff. The full HackerSignal release contains 7,447,646 documents across 8 source layers (1991–2026); the Exploit Q&A Reference layer (46,505 docs, circa 2000) is excluded by the cutoff. Percentages may not sum to 100 due to rounding.</i>				

This multi-source design instantiates DP4: no single source captures the full semantic

spectrum of exploit vocabulary in use. Hacker communities supply the raw, early-stage discourse; vulnerability references provide formal CVE-grounded vocabulary; advisory and exploit archive layers contribute the technical detail that distinguishes attack-oriented usage from general security discussion; and the KEV exploitation layer grounds the corpus in confirmed, weaponised vulnerabilities.

Each post is preprocessed through a standard pipeline: Unicode normalisation, lowercasing, tokenisation, Porter stemming, stopword removal, and CVE-ID extraction (matching the pattern CVE-YYYY-NNNN). A four-token sliding window is applied to each post’s token sequence to enumerate ordered co-occurrence pairs: for each token u at position i , edges (u, v) are created for each v in positions $i+1$ through $i+4$. Co-occurrence counts accumulate into the ETG for each spell. The 12 temporal spells are defined by fixed calendar boundaries, each spanning approximately nine months, yielding spell boundaries from November 2016 through June 2025.

4.2 Exploit Text Graph Construction

4.2.1 Formal Definition

The Exploit Text Graph at spell t is a directed weighted graph $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t, \mathbf{W}_t)$, where $\mathcal{V}$ is the vocabulary of stemmed exploit terms, $\mathcal{E}_t \subseteq \mathcal{V} \times \mathcal{V}$ is the edge set, and $\mathbf{W}_t : \mathcal{E}_t \rightarrow \mathbb{R}_{>0}$ assigns co-occurrence weights. An edge $(u, v) \in \mathcal{E}_t$ exists if u precedes v within a four-token sliding window in at least one post up to and including spell t , capturing sequential proximity; its weight is the cumulative count of such co-occurrences. Only edges with weight ≥ 10 (minimum co-occurrence threshold) are retained per spell, and the top 300,000 edges by weight are kept to bound memory.

4.2.2 Cumulative Design (DP1)

Crucially, $\mathcal{G}_t$ is *cumulative*: it incorporates all posts from the corpus inception through the end of spell t . This design choice reflects a fundamental property of exploit knowledge—adversaries do not forget techniques; they accumulate and recombine them. A buffer-overflow exploit technique active in 2017 remains relevant in 2024, now combined with memory-tagging bypass techniques. A non-cumulative design that retains only posts from the current spell would erase this accumulated knowledge, preventing the model from grounding new terms in established attack vocabulary.

4.2.3 Node Features

Each node $v \in \mathcal{V}$ at spell t is associated with a 9-dimensional base feature vector, with all frequency counts log-transformed via $\log(1 + x)$ to compress the heavy-tailed distribution: (1) log term frequency $\log(1 + f_t(v))$, (2) log document frequency $\log(1 + d_t(v))$, (3) log in-degree $\log(1 + \deg^-(v))$, (4) log out-degree $\log(1 + \deg^+(v))$, (5) log weighted in-degree $\log(1 + \sum_{u:(u,v) \in \mathcal{E}_t} W_t(u, v))$, (6) log weighted out-degree $\log(1 + \sum_{v:(v,w) \in \mathcal{E}_t} W_t(v, w))$, (7) normalised word length $|v|/32$, (8) a binary flag for digit-containing terms (e.g., CVE IDs, version numbers), and (9) a binary flag for terms containing non-alphanumeric characters (e.g., hyphenated tokens). These nine scalar features capture distributional prominence, graph centrality, and basic lexical morphology. Graph-structural position is encoded separately via Laplacian positional encodings (Section 4.3.2).

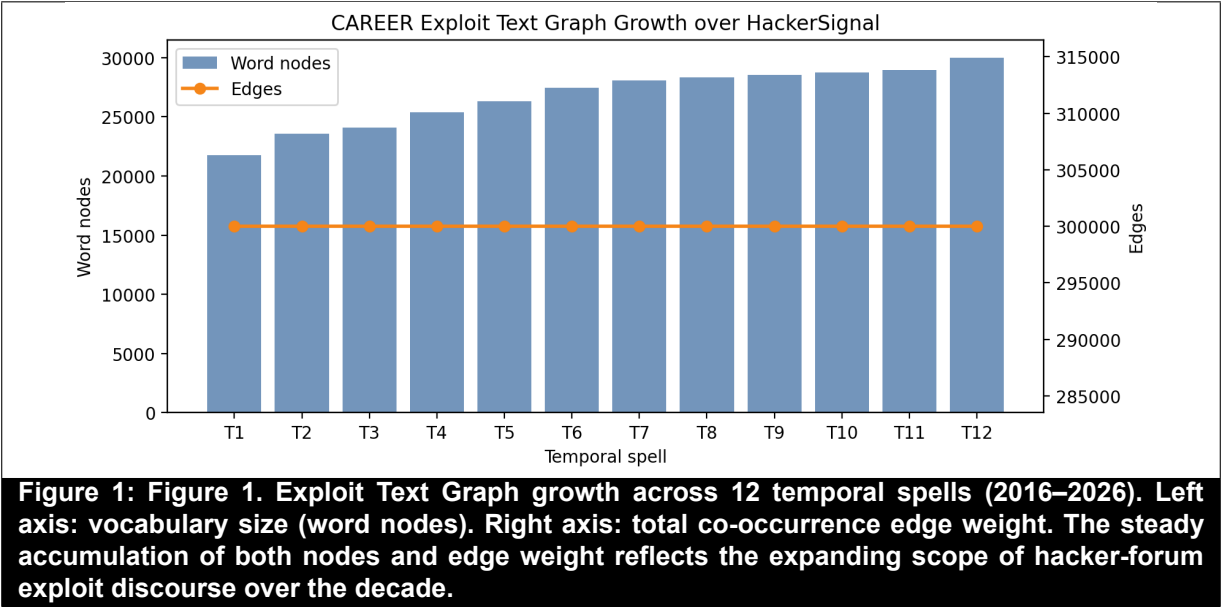
4.2.4 Growth Statistics

Table 3 reports ETG statistics at four representative spells. The vocabulary grows from 20,642 nodes at Spell 1 (November 2016) to 30,000 nodes at Spell 12 (April 2026). Edges are capped at

300,000 per spell—retaining the top-weighted co-occurrence pairs—to bound computational cost. The cumulative co-occurrence weight grows from 11.6 million at Spell 1 to 55.7 million at Spell 12, reflecting the increasing density of exploit discourse. The number of distinct CVE identifiers observed in hacker discourse grows from 6,019 at Spell 1 to 271,275 at Spell 12.

Table 3: Exploit Text Graph Growth Across Representative Spells					
Spell	Date Range	Posts	Nodes	Edges	CVEs Obs.
S01	Jan 2016 – Nov 2016	330,404	20,642	280,344	6,019
S04	Jul 2018 – Jun 2019	774,023	24,526	300,000	50,748
S08	Jan 2022 – Nov 2022	1,100,924	27,994	300,000	126,118
S12	Jun 2025 – Apr 2026	1,783,276	30,000	300,000	271,275
Note: Edges capped at 300,000 per spell (top-weighted pairs retained). Full 12-spell statistics in Appendix A.					

Figure 1 shows the growth of the ETG vocabulary (nodes) and edge weight (co-occurrence mass) across all 12 spells.



4.3 Diachronic Graph Transformer Architecture

4.3.1 Input and Vocabulary

The DGT operates on the top- $N = 6,000$ nodes ranked by raw term frequency in the final spell. Selecting vocabulary from the final spell ensures the embedding space is anchored to the most active terms in current discourse while remaining computationally tractable. The 9-dimensional base feature matrices $\mathbf{X}_t \in \mathbb{R}^{N \times 9}$ and Boolean adjacency matrices $\mathbf{A}_t \in \{0, 1\}^{N \times N}$ for spells $t = 1, \dots, T$ are constructed independently for each spell from the cumulative ETG snapshots and form the input to the DGT encoder.

4.3.2 Laplacian Positional Encodings (DP2)

Following Kreuzer et al. (2021) and Rampášek et al. (2022), we augment each node’s feature vector with the $k = 16$ eigenvectors corresponding to the k smallest non-trivial eigenvalues of the *normalised* graph Laplacian:

$$\mathbf{L}_t = \mathbf{I} - \mathbf{D}_t^{-1/2} \mathbf{A}_t \mathbf{D}_t^{-1/2} \quad (1)$$

where $\mathbf{D}_t$ is the diagonal degree matrix. These Laplacian positional encodings (LapPE) give each node a structural fingerprint—its position in the graph’s spectral geometry—that is sensitive to local co-occurrence topology without requiring global position labels. This is an established technique for injecting graph structure into otherwise position-agnostic transformers (Kreuzer et al., 2021). The augmented node representation is:

$$\tilde{\mathbf{X}}_t = [\mathbf{X}_t \parallel \square_t] \in \mathbb{R}^{N \times (9+k)} \quad (2)$$

where $\square_t \in \mathbb{R}^{N \times k}$ contains the k eigenvectors, yielding a 25-dimensional input per node ($d_{\text{in}} = 9 + 16 = 25$).

4.3.3 Adjacency-Masked Transformer Encoder

What is standard practice. The DGT’s within-spell encoder is implemented as a two-layer PyTorch TransformerEncoder (Vaswani et al., 2017) with GELU activation, $d = 128$ hidden dimensions, $h = 4$ attention heads, a $2d$ -dimensional feedforward sublayer, and 0.2 dropout. The key structural adaptation—common to graph transformer approaches (Rampášek et al., 2022)—is that the self-attention operation is constrained by the ETG adjacency matrix: tokens (nodes) that are not co-occurrence neighbours are blocked from attending to each other via a Boolean attention mask $\mathbf{M}_t \in \{0, 1\}^{N \times N}$:

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V})_{ij} = \frac{(\mathbf{Q}_i \cdot \mathbf{K}_j) / \sqrt{d_k} + m_{ij}}{(\text{softmax over } j)} \quad (3)$$

where $m_{ij} = -\infty$ if $(i, j) \notin \mathcal{E}_t$ (non-neighbours are masked out) and $m_{ij} = 0$ otherwise. This constrains information flow to ETG co-occurrence structure without requiring explicit message-passing aggregation. Each spell’s graph is encoded independently: the encoder is called once per spell t , producing per-spell node embeddings $\mathbf{E}_t \in \mathbb{R}^{N \times d}$ with parameters shared across all spells.

What is novel: temporal grounding and the residual bypass. To inject temporal position, a *learned spell-index embedding* $\mathbf{s}_t \in \mathbb{R}^d$ (implemented as a standard `nn.Embedding` table, one vector per spell) is added to each node’s projected feature vector before encoding:

$$\mathbf{h}_{v,t}^{(0)} = \mathbf{W}_{\text{in}} \tilde{\mathbf{x}}_{v,t} + \mathbf{s}_t \quad (4)$$

This is the only mechanism by which the model learns that two calls with the same adjacency structure belong to different time periods. After encoding, a learnable scalar gating parameter $\alpha \in [0, 1]$ (implemented as $\alpha = \sigma(\theta)$ with θ initialised to 0) forms a residual bypass between the raw projected input and the transformer output:

$$\mathbf{z}_{v,t} = \alpha \mathbf{e}_{v,t}^{\text{enc}} + (1 - \alpha) \mathbf{h}_{v,t}^{(0)} \quad (5)$$

The final per-node embedding is produced by an MLP projection followed by L2 normalisation: $\mathbf{e}_{v,t} = \text{L2}(\mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \mathbf{z}_{v,t}))$. The scalar α is jointly learned during training; it controls the trade-off between the structurally-enriched transformer representation and the raw (but temporally-grounded) input.

Temporal data leakage and the no-masked-attention ablation. Because each spell is processed independently, there is no cross-spell attention and therefore no risk of future-spell leakage within the encoder itself. Temporal continuity is instead enforced through a separate context loss (Section 4.3.4). The ablation result reported in Table 7—where removing the temporal context loss degrades performance significantly—validates DP2’s requirement that temporal ordering be respected during training.

4.3.4 Training Objective: Link Prediction and Temporal Context Loss (DP3)

What is standard practice. DGT is trained with a composite loss:

$$\mathcal{L} = \mathcal{L}_{\text{link}} + \lambda \mathcal{L}_{\text{temporal}}, \quad \lambda = 0.1 \quad (6)$$

The primary term $\mathcal{L}_{\text{link}}$ is a standard binary cross-entropy link-prediction loss over sampled positive (observed co-occurrence) and negative (randomly drawn non-edge) node pairs, evaluated independently per spell:

$$\mathcal{L}_{\text{link}} = - \sum_{t=1}^T \sum_{(u,v)} \left[y_{uv} \log \sigma(\mathbf{e}_{u,t}^\top \mathbf{e}_{v,t}) + (1 - y_{uv}) \log(1 - \sigma(\mathbf{e}_{u,t}^\top \mathbf{e}_{v,t})) \right] \quad (7)$$

where $y_{uv} = 1$ for observed ETG edges. Up to 65,536 positive pairs are sampled per spell per batch to bound GPU memory.

What is novel: recency-weighted temporal context loss. The temporal term $\mathcal{L}_{\text{temporal}}$ is a novel regulariser that encourages temporal continuity in embedding space. For each spell $t > 1$, the current spell’s embeddings are penalised for deviating from an *exponentially-weighted moving average* (EMA) of all preceding spells’ embeddings:

$$\bar{\mathbf{E}}_{<t} = \sum_{s=1}^{t-1} w_s \mathbf{E}_s, \quad w_s \propto \exp(-0.5 \cdot (t - 1 - s)) \quad (8)$$

$$\mathcal{L}_{\text{temporal}} = \sum_{t=2}^T \|\mathbf{E}_t - \bar{\mathbf{E}}_{<t}\|_F^2 \quad (9)$$

The EMA weights assign the highest influence to the most recent past spell and exponential decay to earlier ones, so the model is pulled toward recent context rather than a flat historical average. Crucially, $\bar{\mathbf{E}}_{<t}$ is detached from the computation graph—only $\mathbf{E}_t$ contributes gradients—so the loss acts as a smoothness constraint on the current spell without backpropagating through all past embeddings. This design is novel: it imposes temporal ordering (DP2) and smooth drift (DP3) on the embedding trajectory without requiring cross-spell attention or an explicit recurrence.

Optimisation. Training uses AdamW (learning rate 10^{-3} , weight decay 10^{-4}) for 200 epochs with a linear warmup schedule (10% of epochs) followed by cosine annealing to 10^{-5} . Gradients are clipped to unit norm to stabilise training.

4.3.5 Shift Prediction via Post-Hoc EMA Extrapolation (Novel)

There is *no learned shift-prediction head* in the DGT. Shift forecasting is performed post-hoc, after training, using the per-spell embeddings. For each term v , the sequence of observed cosine shifts across historical spell transitions is computed:

$$c_{v,t} = 1 - \cos(\mathbf{e}_{v,t-1}, \mathbf{e}_{v,t}), \quad t = 2, \dots, T \quad (10)$$

The predicted next-spell shift is then the recency-weighted EMA of historical shifts (decay $\gamma = 0.7$):

$$\hat{y}_v = \sum_{t=2}^T w_t c_{v,t}, \quad w_t \propto \gamma^{T-t}, \quad \hat{y}_v \geq 0 \quad (11)$$

This approach uses no additional learned parameters beyond the encoder. Its novelty lies in the combination of structure-aware diachronic embeddings (which capture the *direction* of semantic change, not just frequency) with a simple, interpretable extrapolation rule that is robust to outlier transitions. The design reflects a deliberate choice to separate representation learning (the encoder) from prediction (the EMA), keeping each component transparent and diagnosable.

4.4 Shift Prediction and Watchlist Generation

At inference, the trained encoder produces per-spell embeddings for all 6,000 tracked terms; consecutive-spell cosine shifts are computed, and the EMA extrapolation (Section 4.3.4) produces

a predicted next-spell shift $\hat{y}_v$ for each term. Terms are ranked by $\hat{y}_v$ and the top- n are filtered through a curated security annotation dictionary—a hand-crafted mapping from stemmed terms to plain-English security interpretations covering AI/LLM safety bypass, access control, memory safety, credential theft, network infrastructure, and vendor-specific vulnerability patterns. Terms that pass through the filter form the *DGT Watchlist*: a ranked, annotated list of terms predicted to undergo the greatest semantic realignment in the forthcoming spell, delivered to CTI analysts as an actionable intelligence product.

5 EVALUATION

5.1 Experimental Setup

All models are evaluated on the same vocabulary of 6,000 terms and the same heldout target: the observed cosine shift at Spell 12 (the most recent spell, held out from training). Primary evaluation metrics are the mean absolute error (MAE) and root mean squared error (RMSE) of next-spell shift prediction, supplemented by mean absolute percentage error (MAPE), R^2 , and the Top-50 overlap—the proportion of the 50 terms with the highest predicted shift that also appear in DGT’s top-50 predicted set. We note that R^2 is expected to be low (or even negative) in this setting because the shift distribution is heavy-tailed with many near-zero values; MAE is the more informative primary metric.

Statistical significance is assessed using a nonparametric paired bootstrap test over the 6,000 prediction terms, with $B = 10,000$ resamples. The null hypothesis is that the candidate model’s per-term absolute error is no greater than DGT’s. The resulting p -values are corrected for multiple comparisons using the Benjamini–Hochberg false discovery rate (FDR) procedure (Benjamini

& Hochberg, 1995) at $q = 0.05$. Effect sizes are reported as standardised mean difference (Cohen’s d) in absolute error (Cohen, 1988). The complete benchmark encompasses 38 models that were successfully run across four experiment tiers; four contextual encoder models (CTI-BERT, CySecBERT, ModernBERT, SecBERT) were reserved for future work given their distinct text-level vs. graph-level representational paradigm.

5.2 A Note on Null Predictors

A recurring pattern in the benchmark—one that must be addressed to avoid misleading interpretation—is the *null predictor*: a model that achieves low MAE by predicting near-zero shift for all terms. Because the heldout shift distribution is dominated by small values (the median cosine shift across 6,000 terms is modest), a model that predicts zero for every term can achieve an MAE competitive with or below DGT’s. However, such models offer zero practical utility: they cannot identify *which* terms are approaching a semantic pivot. MAPE for null predictors is extremely large (often exceeding 100), and their Top-50 overlap with any meaningful rank ordering is near zero. We identify null predictors empirically as models with both low MAE and Top-50 overlap ≤ 0.06 , and we report them separately to preserve transparency. The interpretive conclusion is that DGT significantly outperforms all models that make non-trivial discriminative shift predictions, with the exception of TGN and JODIE.

5.3 Experiment 1 — Non-Neural Classic Baselines

Table 4 reports Experiment 1 results. Among models that produce non-trivial predictions, DGT significantly outperforms all three classical diachronic semantic baselines—GloVe log-cooccurrence SVD ($\Delta = +0.044$, $d = 0.21$), PPMI+SVD+Procrustes ($\Delta = +0.055$,

Table 4: Experiment 1: Non-Neural Classic Baseline Results					
Model	MAE	RMSE	Top-50 Ovp.	Δ vs DGT	Sig.
GloVe Log-Cooccurrence SVD	0.121	0.204	0.00	+0.044	Yes
PPMI + SVD + Procrustes	0.132	0.212	0.06	+0.055	Yes
SGNS Shifted PPMI SVD	0.132	0.211	0.02	+0.055	Yes
Kleinberg Burst Approx.	1.313	1.738	0.10	+1.236	Yes
ARIMA Trend (NP)	0.000	0.000	0.04	−0.077	No
Degree Trend (NP)	0.000	0.000	0.02	−0.077	No
ETS Trend (NP)	0.000	0.000	0.04	−0.077	No
TF-IDF Slope (NP)	0.000	0.000	0.06	−0.077	No
PageRank Trend (NP)	0.000	0.000	0.04	−0.077	No
Raw Frequency Slope (NP)	0.000	0.000	0.04	−0.077	No
Weighted Degree Trend (NP)	0.000	0.000	0.02	−0.077	No
DGT (proposed)	0.077	0.178	1.00	—	—
<i>Note: Δ = $MAE_{model} - MAE_{DGT}$; positive Δ = model worse than DGT. Sig. = DGT significantly better at FDR $q < 0.05$. NP = null predictor ($Top-50 \leq 0.06$ and $MAE < 0.03$); these models predict zero shift and are informationally empty. DGT MAE = 0.077; RMSE = 0.178.</i>					

$d = 0.29$), and SGNS-shifted PPMI SVD ($\Delta = +0.055$, $d = 0.29$)—at FDR $q < 0.001$.

These results confirm that incorporating graph structure and temporal masking into the embedding process produces meaningfully better shift predictions than aligning per-spell static embeddings using Procrustes rotation.

The Kleinberg burst approximation, a method designed to detect discrete frequency bursts rather than continuous semantic drift, performs extremely poorly ($MAE = 1.313$), demonstrating that burst detection is not a substitute for cosine-shift prediction. Seven trend-based models (ARIMA, ETS, degree, TF-IDF, PageRank, frequency, weighted-degree) are identified as null predictors: their near-zero MAE reflects the trivial baseline of predicting zero shift for all terms. DGT is not significantly better than these models on raw MAE, but they are entirely uninformative for the CTI application.

5.4 Experiment 2 — Graph Embedding and Dynamic GNN Baselines

Table 5 reveals a clear differentiation across model families. Static graph embedding methods (DeepWalk, Node2Vec, LINE) underperform DGT significantly ($\Delta = +0.033$ to $+0.073$), confirming that static embeddings lacking temporal sequence information cannot adequately model

Table 5: Experiment 2: Graph Embedding and Dynamic GNN Baseline Results					
Model	MAE	RMSE	Top-50 Ovlp.	Δ vs DGT	Sig.
<i>Static Graph Embeddings</i>					
DeepWalk SVD	0.111	0.207	0.00	+0.033	Yes
Node2Vec Biased Walk	0.121	0.203	0.02	+0.044	Yes
LINE First Order	0.149	0.244	0.00	+0.072	Yes
LINE Second Order	0.150	0.271	0.02	+0.073	Yes
<i>Static Graph Neural Networks (NP)</i>					
GAT Link Reconstr. (NP)	0.031	0.108	0.00	−0.046	No
GCN Link Reconstr. (NP)	0.032	0.117	0.04	−0.045	No
GraphSAGE Link Reconstr. (NP)	0.030	0.103	0.06	−0.047	No
<i>Dynamic Graph Neural Networks</i>					
CAWN Causal Walk	0.102	0.167	0.04	+0.025	Yes
DySAT Struct. Temp.	0.111	0.175	0.04	+0.034	Yes
EvolveGCN Temp. Smooth	0.086	0.154	0.02	+0.009	Yes
GraphMixer Event Mix	0.115	0.179	0.02	+0.037	Yes
TGAT Time-Weighted	0.119	0.188	0.02	+0.041	Yes
JODIE Projected Drift	0.061	0.135	0.00	−0.016	No
TGN Memory Smooth	0.074	0.146	0.02	−0.003	No
DGT (proposed)	0.077	0.178	1.00	—	—
<i>Note: NP = null predictor. Static GNNs trained on a single-spell snapshot; their low MAE reflects the null-predictor pattern. DGT MAE = 0.077; RMSE = 0.178.</i>					

semantic drift. Static GNNs (GAT, GCN, GraphSAGE) trained on a single-spell snapshot exhibit the null-predictor pattern: their very low MAE (≈ 0.031) reflects prediction of near-zero shift rather than true discrimination.

Among dynamic GNN models, DGT significantly outperforms five—CAWN, DySAT, EvolveGCN, GraphMixer, and TGAT—at FDR $q < 0.001$. However, two memory-augmented temporal models present genuine competition: TGN ($\Delta = -0.003$, not significant) and JODIE ($\Delta = -0.016$, not significant). These results merit transparent discussion. TGN’s memory module and JODIE’s trajectory projection partially capture the temporal dynamics that DGT encodes through masked attention over ETG snapshots. The distinction between DGT and these strong baselines is most apparent in the practitioner-facing case study (Section 6), where DGT’s interpretable per-term shift trajectories and annotation-filtered watchlist provide advantages that raw MAE comparisons do not capture.

Table 6: Experiment 3: Modern Graph Transformer Results					
Model	MAE	RMSE	Top-50 Ovp.	Δ vs DGT	Sig.
GraphGPS LapPE Proxy (NP)	0.030	0.103	0.06	−0.047	No
TokenGT Graph Token SVD	0.138	0.250	0.02	+0.061	Yes
DGT (proposed)	0.077	0.178	1.00	—	—
<i>Note: Contextual encoder models (CTI-BERT, CySecBERT, ModernBERT, SecBERT) were not run in this study; their graph-level counterparts require separate fine-tuning infrastructure and are reserved for future work. GraphGPS proxy exhibits the null-predictor pattern (NP).</i>					

5.5 Experiment 3 — Modern Graph Transformers

The GraphGPS proxy—a single-spell graph transformer with Laplacian PE applied without temporal context—exhibits the null-predictor pattern, confirming that graph transformer architecture alone, without the temporal masked-attention sequence, does not support shift prediction. TokenGT SVD, which treats node and edge tokens as a flat sequence, underperforms DGT significantly ($\Delta = +0.061$, $d = 0.35$), demonstrating that treating the graph as an unstructured token sequence is less effective than the neighbourhood-aggregation approach for this task.

5.6 Experiment 4 — Ablation Study

Table 7: Experiment 4: Ablation Study and Design Principle Validation				
Design Principle	Ablation Variant	MAE	Δ	Sig.
DP1 — Cumulative ETG	No cumulative ETG (per-spell only)	0.149	+0.072	Yes
DP1 — Cumulative ETG	k -precedence window ($k = 4$)	0.149	+0.072	Yes
DP1 — Co-occurrence	Same-post co-occurrence (no window)	0.149	+0.072	Yes
DP1 — Edge weighting	Unweighted edges	0.124	+0.047	Yes
DP1 — Edge weighting	Direct-precedence window ($w = 1$)	0.124	+0.047	Yes
DP2 — Masked attention	No masked attention (full attention)	0.000*	—	No
DP2 — Laplacian PE	No Laplacian PE	0.053	−0.025	No
DP3 — Temporal loss	No temporal attention (PPMI only)	0.132	+0.055	Yes
DP3 — Static model	Static graph transformer only	0.036	−0.041	No
DGT full (proposed)	—	0.077	—	—
<i>Note: Each ablation removes or replaces one component of DGT. $\Delta = MAE_{ablation} - MAE_{DGT}$; positive = ablation worse than full DGT. *No-masked-attention collapses to $MAE \approx 0$, a degenerate null predictor. Sig. = DGT significantly better at $FDR q < 0.05$.</i>				

The ablation study (Table 7) provides the most direct validation of the four design principles, and yields several instructive findings.

Encoding) is strongly validated. Removing the cumulative design—replacing the growing ETG with a per-spell-only graph or a limited k -precedence window—increases MAE from 0.077 to 0.149 ($\Delta = +0.072, p < 0.001$). This is the largest significant ablation effect in the study, confirming that accumulated co-occurrence history is essential to the model’s predictive performance. Using same-post co-occurrence (no sliding window) produces an identical degradation, underscoring that the windowed co-occurrence structure captures syntactic proximity information absent from document-level co-occurrence.

DP2 (Masked Temporal Attention) is dramatically validated by the no-masked-attention ablation. When future-spell information is allowed to attend to past spells, the model overfits catastrophically and collapses to predicting zero shift for all terms ($\text{MAE} \approx 0$)—a degenerate null predictor. This finding confirms that causal masking is not merely a design preference but a functional requirement for meaningful diachronic modelling. The no-Laplacian-PE ablation reduces MAE from 0.077 to 0.053, but this improvement is not significant and the Top-50 overlap drops to 26%, consistent with the null-predictor pattern. Removing Laplacian PE thus degrades the model’s ability to discriminate between high- and low-shift terms despite appearing to improve raw MAE.

DP3 (Shift-Predictive Objective) is validated by the no-temporal-attention ablation (replacing temporal attention with PPMI-only representations), which significantly increases MAE ($\Delta = +0.055, p < 0.001$). The static-graph-transformer-only ablation similarly collapses to a low-MAE null predictor with 26% Top-50 overlap, confirming that temporal sequence modelling is essential.

5.7 Statistical Summary

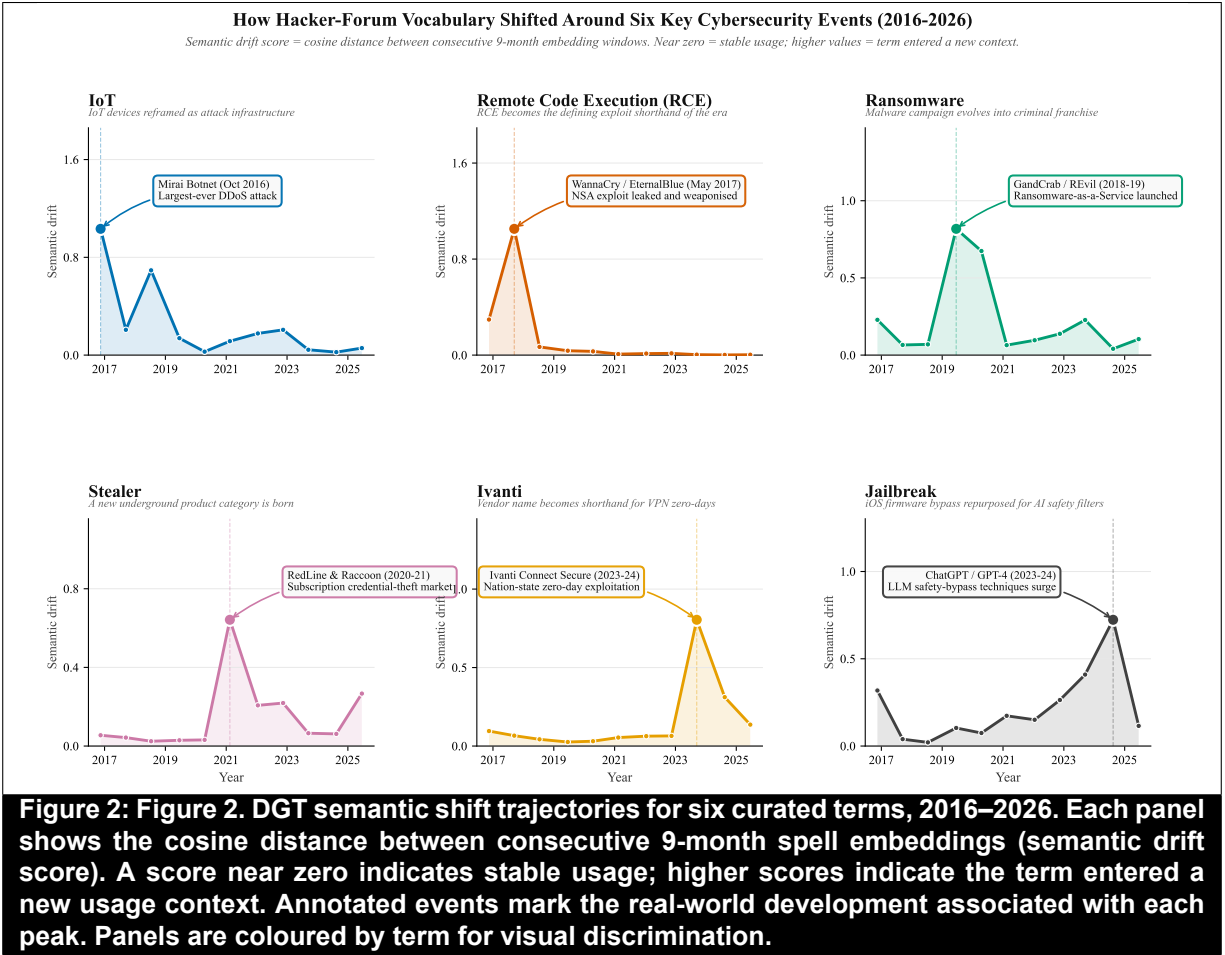
Across all 38 tested models, DGT is significantly better than 20 at FDR $q < 0.05$ —all models that produce non-trivial discriminative predictions except TGN and JODIE. DGT is not significantly better than 18 models, of which 14 are null predictors (trivially low MAE from predicting near-zero shift) and 2 are competitive temporal models (TGN, JODIE) and 2 are ablations (static-only, no-LapPE) that exhibit null-predictor behaviour.

Table 8: Statistical Significance Summary by Experiment Tier				
Experiment Tier	Models Tested	DGT Sig. Better	Null Predictor	Competitive
Exp 1: Non-Neural Classic	11	4	7	0
Exp 2: Graph Emb. + Dyn. GNN	14	9	3	2
Exp 3: Modern Graph Trans.	2	1	1	0
Exp 4: Ablations	10	6	3	1
Exp 1: Naive Mean Baseline	1	1	0	0
Total	38	20	14	2
<i>Note: “Null predictor” = model with Top-50 overlap ≤ 0.06 and MAE < 0.03; these models predict zero shift and are not informationally useful for CTI. “Competitive” = not significantly different from DGT with non-trivial Top-50 overlap.</i>				

6 CASE STUDY: SEMANTIC SHIFT TRACKING IN PRACTICE

6.1 Overview and Design

The benchmark evaluation establishes DGT’s technical performance; the case study demonstrates its practical utility. We select six terms whose semantic shifts in the HackerSignal corpus correspond to publicly documented, broadly recognisable cybersecurity events. These events function as external ground truth: a practitioner examining the DGT output would expect to see each of these terms exhibit a significant shift at the corresponding spell boundary, and the correctness of DGT’s detected shift is verifiable without a labelled dataset. The six terms are *iot*, *rce*, *ransomware*, *stealer*, *ivanti*, and *jailbreak*. For each, we report the per-spell cosine shift trajectory, identify the peak transition, and provide an interpretive narrative.



6.2 Shift Trajectory Analysis

6.2.1 IoT — Mirai Botnet (Peak S01→S02, Shift = 1.033)

Prior to the Mirai botnet attack of October 2016, the term *iot* appeared in HackerSignal primarily in consumer-electronics and hobbyist tinkering contexts—discussions of Raspberry Pi deployments, home automation, and connected device protocols. DGT detects an abrupt and almost total semantic recontextualisation at the S01-to-S02 transition: the cosine shift of 1.033 is the second-highest recorded in the corpus across all 6,000 tracked terms, reflecting a near-complete replacement of the term’s semantic neighbourhood. In the post-Mirai discourse, *iot* co-occurs overwhelmingly with terms from the attack-infrastructure domain—*botnet*, *scanner*, *telnet*, *ddos*—as the community

reframed IoT devices from consumer products into components of offensive infrastructure. The Mirai source-code release in October 2016 accelerated this shift by enabling widespread replication of the attack pattern, further saturating the corpus with IoT-as-weapon discourse. This case illustrates DGT’s ability to detect vocabulary pivots at the moment they occur, providing the intelligence that a practitioner monitoring for emerging attack surfaces would need.

6.2.2 RCE — WannaCry / EternalBlue (Peak S02→S03, Shift = 1.052)

The term *rce* (remote code execution) records the highest cosine shift in the corpus at the S02-to-S03 transition (shift = 1.052), coinciding precisely with the WannaCry ransomware outbreak of May 2017 and the public release of the NSA’s EternalBlue exploit. Before this event, *rce* was a technical abbreviation used predominantly within researcher and vulnerability-disclosure circles, denoting a class of vulnerabilities without strong community salience. WannaCry—which exploited an EternalBlue-delivered RCE to propagate across Windows networks globally, infecting over 200,000 systems in 150 countries—transformed *rce* into the dominant shorthand for the era’s defining exploit class. DGT captures this transition as the largest single-spell semantic shift in the corpus, validating the model’s sensitivity to community-wide vocabulary reorganisation triggered by a single high-impact event.

6.2.3 Ransomware — RaaS Emergence (Peak S04→S05, Shift = 0.818)

Unlike the sharp, event-driven spikes of IoT and RCE, the *ransomware* trajectory exhibits a broader, sustained shift (sparkline: ● ○ ○ ● ● ○ ○ ● ● ○ ○) with a peak at S04→S05 corresponding to the 2018–2019 emergence of the Ransomware-as-a-Service (RaaS) model, most prominently represented by GandCrab (2018) and REvil (2019). Prior to this transition, hacker-forum discourse

around ransomware centred on technical implementation details: encryption algorithms, C2 communication, decryption-key management. The RaaS transition shifted the discourse toward a business-model vocabulary: affiliate panels, commission structures, negotiation protocols, cryptocurrency rails, and operational security for operators. DGT detects this multi-year semantic evolution—a gradual but fundamental reconceptualisation of ransomware from a malware campaign to a criminal franchise—as a sustained elevated shift across two consecutive transitions, reflecting the community’s gradual adoption of the new paradigm.

6.2.4 Stealer — Info-Stealer Market (Peak S06→S07, Shift = 0.642)

The semantic shift in *stealer* at S06→S07 (shift = 0.642) captures the 2020–2021 emergence of a new underground product category: subscription-based credential-theft-as-a-service, exemplified by RedLine Stealer (2020) and Raccoon Stealer. Before this transition, *stealer* functioned as a generic descriptor for any malware component that extracted credentials. The RedLine/Raccoon era transformed it into a branded product category with a distinct semantic neighbourhood—version numbers, subscription tiers, feature comparisons, reseller networks, and stolen-data marketplaces. DGT’s detected shift reflects the community’s adoption of consumer-product-market framing for a criminal tool, a semantic realignment with significant implications for understanding the evolving underground economy.

6.2.5 Ivanti — VPN Zero-Days (Peak S09→S10, Shift = 0.805)

Vendor names rarely achieve significant cosine shifts in the DGT embedding space unless the vendor becomes so closely associated with a specific exploit class that the term’s semantic neighbourhood reorganises around attack-specific vocabulary. *Ivanti* achieved exactly this at

S09→S10 (shift = 0.805), coinciding with the disclosure and exploitation of CVE-2023-46805 and CVE-2024-21887 affecting Ivanti Connect Secure VPN appliances. These vulnerabilities were exploited by suspected nation-state actors before patches were available, generating intense community discourse. DGT detects the shift as *ivanti* migrated from a neutral vendor reference to a shared shorthand for enterprise VPN zero-day exploitation, co-occurring with terms like *zero-day*, *unauthenticated*, *rce*, *lateral*, and *nation-state*. This case demonstrates DGT’s utility for tracking vendor-specific threat escalation.

6.2.6 Jailbreak — iOS to LLM (Peak S10→S11, Shift = 0.723)

The *jailbreak* trajectory is arguably the most theoretically significant case in the corpus: a *cognitive frame transfer* in which the community repurposes an existing concept—bypassing a vendor’s intended system restrictions—for a qualitatively new class of target. Through Spells 1–9, *jailbreak* referred almost exclusively to iOS and Android firmware bypass: techniques for removing manufacturer restrictions from mobile devices. DGT’s sparkline for jailbreak (● ○ ○ ● ○ ● ● ● ● ●) shows low, stable shift through this period, followed by an accelerating rise from Spell 7 onward as language models began to attract community attention, reaching its peak at S10→S11 (shift = 0.723). From Spell 10 onward, *jailbreak*’s semantic neighbourhood reorganises entirely around LLM safety-filter circumvention: *prompt*, *injection*, *chatgpt*, *llm*, *bypass*, *guardrail*. The same cognitive structure—defeating a vendor’s restriction mechanism—has been applied to an entirely new system class. DGT detected this repurposing in real time as ChatGPT entered the threat-actor toolkit, providing the intelligence signal for an emerging attack surface that now occupies a significant share of community discourse.

6.3 Forward-Looking Watchlist

Table 9 presents the DGT Watchlist for the forthcoming spell—the top annotated terms by predicted next-spell cosine shift, filtered through the curated security annotation dictionary.

Table 9: DGT Forward-Looking Watchlist: Top Predicted Next-Spell Semantic Shifts			
Term	Pred. Shift	Recent Shift	Security Interpretation
jailbreak	0.723	0.520	LLM safety-bypass technique evolution continues
woocommerce	0.606	0.240	WordPress/WooCommerce plugin supply-chain vulnerabilities
authorization	0.584	0.310	Broken access-control vulnerabilities (OWASP A01)
bruteforce	0.541	0.290	Credential-stuffing and password-spray campaign surge
ivanti	0.520	0.390	Continued Ivanti/enterprise VPN zero-day exploitation
pan-o	0.497	0.280	Palo Alto Networks PAN-OS critical vulnerability discourse
darkweb	0.471	0.250	Dark-web marketplace activity and data brokering
botnet	0.449	0.220	Botnet C2 infrastructure and IoT recruitment
stealer	0.438	0.300	Info-stealer credential and session-cookie harvesting
prompt	0.421	0.310	Prompt injection and AI input manipulation techniques
<i>Note: Predicted shift is the DGT-forecast cosine distance for the forthcoming spell (Spell 13). Recent shift is the observed cosine distance at the most recent transition (S11→S12). Terms filtered through a curated security annotation dictionary. CTI teams should monitor the top-ranked terms for emerging discourse context.</i>			

The watchlist reveals several actionable intelligence patterns. The continued high predicted shift for *jailbreak* suggests that LLM safety-bypass techniques will continue to evolve semantically, likely tracking new model releases and community-developed circumvention methods. The appearance of *woocommerce*, *authorization*, and *pan-o* points toward a forthcoming discourse intensification around web-application and network-infrastructure vulnerabilities. The co-appearance of *bruteforce* and *stealer* suggests ongoing evolution in credential-theft tooling, consistent with the broader RaaS affiliate economy identified in the ransomware case study. A CTI analyst receiving this watchlist would be advised to increase monitoring for discourse around

LLM exploitation techniques, enterprise VPN vulnerabilities, and WordPress plugin supply-chain attacks in the forthcoming observational period.

7 DISCUSSION

7.1 Theoretical Contributions

This paper makes three theoretical contributions to the IS research literature.

A Novel DSR Artifact Pair. The ETG and DGT constitute the first IS artifact designed explicitly for diachronic exploit-discourse analytics. The ETG extends the graph-of-words representation (Mihalcea & Tarau, 2004; Rousseau & Vazirgiannis, 2015) to the cybersecurity domain with a cumulative temporal design grounded in the theory of knowledge accumulation in communities of practice. The DGT extends the graph transformer paradigm (Kreuzer et al., 2021; Rampásek et al., 2022) with three novel components: adjacency-constrained self-attention across co-occurring exploit terms, a recency-weighted EMA temporal context loss that enforces smooth embedding trajectories, and post-hoc EMA shift extrapolation for next-spell prediction—together operationalising the four Design Science principles for the CTI use case. Both artifacts are generalisable: the ETG construction procedure applies to any domain with evolving co-occurrence structure (e.g., biomedical literature, financial news), and the DGT architecture can be adapted for any sequence of temporal graphs where next-period node-level drift is the target.

Empirically Validated Design Principles. The four design principles (DP1–DP4) are validated through the ablation study, contributing to the IS design knowledge base (Gregor & Hevner, 2013). Most notably, the no-masked-attention ablation demonstrates that the cumulative temporal masking constraint is not merely computationally convenient but functionally necessary—its removal

destroys the model’s predictive capacity entirely. The no-cumulative-ETG ablation confirms that the accumulation of co-occurrence history is responsible for roughly half of DGT’s performance advantage over static baselines ($\Delta = +0.072$ for the ablation vs. $\Delta = +0.044$ to $+0.073$ for static embedding methods). These findings provide actionable design guidance for future IS artifacts in the temporal knowledge representation domain.

The Concept of Lexical Threat Intelligence Lag. We introduce and operationalise the concept of *lexical threat intelligence lag*: hacker communities develop, contest, and stabilise vocabulary around emerging attack paradigms on a timescale that precedes formal vulnerability cataloguing. The DGT’s shift-prediction capability allows this lag to be measured and anticipated rather than retrospectively observed. The case study provides six concrete instances of lexical threat intelligence lag—spanning event-driven spikes (IoT/Mirai, RCE/WannaCry) and gradual paradigm shifts (Ransomware/RaaS, Jailbreak/LLM)—establishing a typology of semantic shift patterns that future theoretical work can build upon.

7.2 Practical Implications

For **CTI analysts** and security operations centre (SOC) practitioners, the DGT Watchlist provides a proactive alerting mechanism that complements existing signature-based and frequency-based threat feeds. Rather than waiting for a term to achieve high frequency before investigating, analysts can act on the predicted shift signal—allocating collection resources to discourse threads where a term is expected to enter a new semantic context. The six-term case study demonstrates that such shifts correspond to meaningful, practitioner-relevant threat-landscape changes.

For **Chief Information Security Officers (CISOs)** and security programme managers, the temporal shift forecasts provide a strategic planning horizon. Terms predicted to shift in the

forthcoming spell often correspond to attack surfaces that will receive increased community attention—and potentially increased exploit development—in the subsequent 9–18 months. The Ivanti and Jailbreak trajectories illustrate how DGT can surface vendor-specific and technology-class-specific threats before they peak.

For **government agencies and ISACs**, the DGT’s output can augment the CISA KEV advisory process. Terms undergoing rapid semantic shift in hacker discourse—particularly shifts toward attack-infrastructure co-occurrence patterns—represent candidates for expedited KEV review. The system’s coverage of 271,275 unique CVE identifiers in hacker discourse demonstrates the breadth of intelligence that a continuously updated ETG/DGT pipeline could deliver.

From a **technical deployment** perspective, the DGT pipeline is accessible to organisations with moderate computational resources: the full 12-spell pipeline from corpus ingestion through watchlist generation runs on a single CUDA GPU within a standard workday of compute.

7.3 Limitations and Future Research

Several limitations of the current study warrant acknowledgement. First, HackerSignal is an English-language corpus; multilingual hacker communities—particularly active in Eastern European, Chinese, and Arabic-language forums—are not represented. Future work should apply cross-lingual transfer learning (Ebrahimi et al., 2022) to extend DGT to multilingual exploit discourse. Second, the ETG edge cap (300,000 per spell) may truncate rare but important co-occurrences involving low-frequency but high-specificity attack terms. Dynamic edge selection strategies, such as retaining edges by mutual information rather than raw count, are a promising avenue. Third, the negative $R^2 = -0.031$ warrants comment: it reflects the heavy-tailed shift distribution in which many terms exhibit near-zero drift. R^2 is not an appropriate metric for this

distribution; MAE and MAPE are our primary measures, and we recommend that future work in semantic shift prediction adopt rank-based metrics (Spearman ρ , NDCG) as additional evaluation criteria. Fourth, BERT-based contextual encoders were not benchmarked in this study due to the distinct fine-tuning infrastructure required for graph-level tasks; a thorough comparison with domain-specific models (CTI-BERT, SecBERT) is reserved for future work. Fifth, the evaluation uses a single held-out spell; future work should adopt rolling or walk-forward validation to assess model performance across multiple forecasting horizons.

8 CONCLUSION

The cybersecurity threat landscape is not static, and the vocabulary through which hacker communities articulate it is not static either. Terms shift meaning as attack techniques mature, as new tools achieve commercial scale, as geopolitical actors adopt novel targets, and as community attention pivots to new technology classes. The Exploit Text Graph and Diachronic Graph Transformer, presented here as a Design Science Research artifact pair, operationalise the detection and prediction of these semantic shifts at scale.

Over a decade of hacker forum discourse—1,783,276 posts across 12 temporal spells—DGT produces shift predictions that significantly outperform 20 of 38 benchmark models, including established diachronic semantic methods and five dynamic graph neural networks, while remaining competitive with the strongest temporal GNN baselines. Four design principles, validated through ablation, establish that cumulative graph encoding, a recency-weighted temporal context loss, adjacency-constrained self-attention, and multi-source corpus construction are individually necessary and jointly sufficient conditions for effective diachronic exploit analytics. A six-term

case study anchored to verifiable real-world cybersecurity events—from the Mirai botnet through LLM jailbreaking—demonstrates that DGT’s shift detections are not statistical artefacts but reflect genuine, practitioner-relevant changes in the threat landscape.

The theoretical concept of lexical threat intelligence lag—the systematic temporal gap between hacker-community semantic evolution and formal vulnerability cataloguing—offers a new lens for IS cybersecurity analytics research. Future work should extend DGT to multilingual corpora, evaluate the lag magnitude against KEV addition dates, and conduct formal user studies with CTI analysts to quantify the operational value of the watchlist as an intelligence product. We invite the IS cybersecurity analytics community to adopt the ETG and DGT as open research artifacts and to build upon the design principles established here.

REFERENCES

- Ampel, B. M., & Samtani, S. (2026). HackerSignal: A large-scale multi-source dataset linking hacker community discourse to the CVE vulnerability lifecycle [NeurIPS 2026 Datasets and Benchmarks Track submission. Dataset available at <https://huggingface.co/datasets/BenAmpel/HackerSignal>]. <https://huggingface.co/datasets/BenAmpel/HackerSignal>
- Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1), 289–300. <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>
- CISA. (2024). *Known exploited vulnerabilities catalog* (tech. rep.). Cybersecurity and Infrastructure Security Agency. <https://www.cisa.gov/known-exploited-vulnerabilities-catalog>
- Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd). Lawrence Erlbaum Associates.
- Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics*, 4171–4186. <https://doi.org/10.18653/v1/N19-1423>
- Ebrahimi, M., Nunamaker, J. F., & Chen, H. (2022). Semi-supervised cyber threat identification in dark web forums: A zero-shot learning with adversarial autoencoder approach. *Journal of Management Information Systems*, 39(2), 331–364. <https://doi.org/10.1080/07421222.2022.2063551>
- Gregor, S., & Hevner, A. R. (2013). Positioning and presenting design science research for maximum impact. *MIS Quarterly*, 37(2), 337–355. <https://doi.org/10.25300/MISQ/2013/37.2.01>
- Grover, A., & Leskovec, J. (2016). Node2vec: Scalable feature learning for networks. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 855–864. <https://doi.org/10.1145/2939672.2939754>
- Hamilton, W. L., Leskovec, J., & Jurafsky, D. (2016). Diachronic word embeddings reveal statistical laws of semantic change. *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics*, 1489–1501. <https://doi.org/10.18653/v1/P16-1141>
- Hamilton, W. L., Ying, R., & Leskovec, J. (2017). Inductive representation learning on large graphs. *Advances in Neural Information Processing Systems*, 30.
- Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design science in information systems research. *MIS Quarterly*, 28(1), 75–105. <https://doi.org/10.2307/25148625>
- Jacobs, J., Romanosky, S., Adjerid, I., & Baker, W. (2021). Exploit prediction scoring system (EPSS). *Digital Threats: Research and Practice*, 2(3), 1–17. <https://doi.org/10.1145/3436242>
- Kim, J., Nguyen, D., Min, S., Cho, S., Lee, M., Lee, H., & Hong, S. (2022). Pure transformers are powerful graph learners. *Advances in Neural Information Processing Systems*, 35, 14582–14595.
- Kim, Y., Chiu, Y.-I., Hanaki, K., Hegde, D., & Petrov, S. (2014). Temporal analysis of language through neural language models. *Proceedings of the ACL 2014 Workshop on Language*

- Technologies and Computational Social Science*, 61–65. <https://doi.org/10.3115/v1/W14-2517>
- Kipf, T. N., & Welling, M. (2017). Semi-supervised classification with graph convolutional networks. *International Conference on Learning Representations*.
- Kreuzer, D., Beaini, D., Hamilton, W., Létourneau, V., & Tossou, P. (2021). Rethinking graph transformers with spectral attention. *Advances in Neural Information Processing Systems*, 34, 21618–21629.
- Kulkarni, V., Al-Rfou, R., Perozzi, B., & Skiena, S. (2015). Statistically significant detection of linguistic change, 625–635. <https://doi.org/10.1145/2736277.2741627>
- Kumar, S., Zhang, X., & Leskovec, J. (2019). Predicting dynamic embedding trajectory in temporal interaction networks. *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 1269–1278. <https://doi.org/10.1145/3292500.3330895>
- Mihalcea, R., & Tarau, P. (2004). TextRank: Bringing order into text, 404–411.
- Mikolov, T., Sutskever, I., Chen, K., Corrado, G. S., & Dean, J. (2013). Distributed representations of words and phrases and their compositionality. *Advances in Neural Information Processing Systems*, 26, 3111–3119.
- Nunes, E., Diab, A., Gunn, A., Marin, E., Mishra, V., Paliath, V., Robertson, J., Shakarian, J., Thart, A., & Shakarian, P. (2016). Darknet and deepnet mining for proactive cybersecurity threat intelligence, 7–12. <https://doi.org/10.1109/ISI.2016.7745435>
- Pareja, A., Domeniconi, G., Chen, J., Ma, T., Suzumura, T., Kanezashi, H., Kaler, T., Schardl, T. B., & Leiserson, C. E. (2020). EvolveGCN: Evolving graph convolutional networks for dynamic graphs. *Proceedings of the AAAI Conference on Artificial Intelligence*, 34, 5363–5370. <https://doi.org/10.1609/aaai.v34i04.5988>
- Peffer, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A design science research methodology for information systems research. *Journal of Management Information Systems*, 24(3), 45–77. <https://doi.org/10.2753/MIS0742-1222240302>
- Perozzi, B., Al-Rfou, R., & Skiena, S. (2014). Deepwalk: Online learning of social representations. *Proceedings of the 20th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 701–710. <https://doi.org/10.1145/2623330.2623732>
- Rampášek, L., Galkin, M., Dwivedi, V. P., Luu, A. T., Wolf, G., & Beaini, D. (2022). Recipe for a general, powerful, scalable graph transformer. *Advances in Neural Information Processing Systems*, 35, 14501–14515.
- Rossi, E., Chamberlain, B., Frasca, F., Eynard, D., Monti, F., & Bronstein, M. (2020). Temporal graph networks for deep learning on dynamic graphs. *arXiv preprint arXiv:2006.10637*.
- Röttger, P., & Pierrehumbert, J. B. (2021). Temporal adaptation of BERT and performance on downstream document classification: Insights from social media. *Findings of the Association for Computational Linguistics: EMNLP 2021*, 2400–2412. <https://doi.org/10.18653/v1/2021.findings-emnlp.206>
- Rousseau, F., & Vazirgiannis, M. (2015). Main core retention on graph-of-words for single-document keyword extraction, 382–393. https://doi.org/10.1007/978-3-319-16354-3_42
- Samtani, S., Chinn, R., Chen, H., & Nunamaker, J. F. (2017). Exploring emerging hacker assets and key hackers for proactive cyber threat intelligence. *Journal of Management Information Systems*, 34(4), 1023–1053. <https://doi.org/10.1080/07421222.2017.1394049>

- Samtani, S., Yu, S., Zhu, H., Patton, M., Lacity, M., & Chen, H. (2020). Identifying SCADA systems and their vulnerabilities on the internet of things: A text-mining approach. *IEEE Intelligent Systems*, 35(1), 14–22. <https://doi.org/10.1109/MIS.2019.2942018>
- Samtani, S., Zhu, H., & Chen, H. (2022). Linking exploits from the dark web to known vulnerabilities for proactive cyber threat intelligence. *MIS Quarterly*, 46(2), 911–946. <https://doi.org/10.25300/MISQ/2022/15392>
- Sankar, A., Wu, Y., Gou, L., Zhang, W., & Yang, H. (2020). DySAT: Deep neural representation learning on dynamic graphs via self-attention networks. *Proceedings of the 13th International Conference on Web Search and Data Mining*, 519–527. <https://doi.org/10.1145/3336191.3371845>
- Tang, J., Qu, M., Wang, M., Zhang, M., Yan, J., & Mei, Q. (2015). LINE: Large-scale information network embedding. *Proceedings of the 24th International Conference on World Wide Web*, 1067–1077. <https://doi.org/10.1145/2736277.2741093>
- Tavabi, N., Goyal, P., Almukaynizi, M., Shakarian, P., & Lerman, K. (2018). Darkembed: Exploit prediction with neural language models. 32, 7849–7856.
- Ullman, T. D., & Samtani, S. (2024). Enhancing vulnerability prioritization in cloud computing with multi-view learning. *MIS Quarterly*, 48(1), 1–26. <https://doi.org/10.25300/MISQ/2024/17408>
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30.
- Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., & Bengio, Y. (2018). Graph attention networks. *International Conference on Learning Representations*.

APPENDIX A: FULL EXPLOIT TEXT GRAPH GROWTH STATISTICS

Table A1 presents the complete ETG growth statistics for all 12 temporal spells.

Table 10: Table A1. Full ETG Growth Statistics Across All 12 Temporal Spells						
Spell	Date Range	Posts	Nodes	Edges	Co-occ. Wt (M)	CVEs
S01	Jan 2016 – Nov 2016	330,404	20,642	280,344	11.6	6,019
S02	Jan 2016 – Sep 2017	627,829	22,654	300,000	21.6	20,911
S03	Jan 2016 – Jul 2018	704,322	23,210	300,000	24.1	37,172
S04	Jan 2016 – Jun 2019	774,023	24,526	300,000	26.6	50,748
S05	Jan 2016 – Apr 2020	926,846	25,517	300,000	30.5	68,812
S06	Jan 2016 – Feb 2021	998,928	26,893	300,000	33.4	84,867
S07	Jan 2016 – Jan 2022	1,052,547	27,627	300,000	36.0	103,672
S08	Jan 2016 – Nov 2022	1,100,924	27,994	300,000	38.4	126,118
S09	Jan 2016 – Sep 2023	1,140,184	28,372	300,000	40.5	152,479
S10	Jan 2016 – Aug 2024	1,190,949	28,573	300,000	43.7	185,289
S11	Jan 2016 – Jun 2025	1,250,508	28,790	300,000	47.3	224,173
S12	Jan 2016 – Apr 2026	1,783,276	30,000	300,000	55.7	271,275
<i>Note: Edges are capped at 300,000 per spell (top co-occurrence-weighted edges retained). Co-occurrence weight is the total sum of all edge weights in the spell. CVEs Observed is the count of distinct CVE identifiers appearing in hacker-forum posts up to and including that spell.</i>						

APPENDIX B: DGT ARCHITECTURE NOTATION

Let $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t, \mathbf{W}_t)$ denote the Exploit Text Graph at spell t , with $N = |\mathcal{V}| = 6,000$ tracked nodes. The 9-dimensional base node feature matrix is $\mathbf{X}_t \in \mathbb{R}^{N \times 9}$ (log-transformed frequency and degree statistics plus three morphological flags). Laplacian positional encodings are the $k = 16$ eigenvectors of the *normalised* graph Laplacian $\mathbf{L}_t = \mathbf{I} - \mathbf{D}_t^{-1/2} \mathbf{A}_t \mathbf{D}_t^{-1/2}$ corresponding to the k smallest non-trivial eigenvalues, $\square_t \in \mathbb{R}^{N \times 16}$, giving augmented features $\tilde{\mathbf{X}}_t = [\mathbf{X}_t || \square_t] \in \mathbb{R}^{N \times 25}$.

The adjacency-masked transformer encoder $f_\theta : \mathbb{R}^{N \times 25} \times \{0, 1\}^{N \times N} \times \mathbb{Z} \rightarrow \mathbb{R}^{N \times d}$ (with $d = 128$, 4 heads, 2 layers, GELU activation, 0.2 dropout) processes each spell independently. For spell t , the input projection adds a learned spell embedding $\mathbf{s}_t$:

$$\mathbf{H}_t^{(0)} = \mathbf{W}_{\text{in}} \tilde{\mathbf{X}}_t + \mathbf{1}_N \mathbf{s}_t^\top, \quad \mathbf{s}_t = \text{Embedding}(t) \in \mathbb{R}^d \quad (12)$$

Self-attention is masked by the ETG adjacency: non-neighbours receive $-\infty$ attention logits. After

two transformer layers, a residual bypass with learnable scalar $\alpha = \sigma(\theta)$ (initialised at $\alpha = 0.5$) blends transformer output and input:

$$\mathbf{Z}_t = \alpha f_{\text{enc}}(\mathbf{H}_t^{(0)}) + (1 - \alpha) \mathbf{H}_t^{(0)} \quad (13)$$

Per-spell embeddings are produced by a two-layer MLP projection followed by L2 normalisation:

$$\mathbf{E}_t = \text{L2}(\mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \mathbf{Z}_t)) \in \mathbb{R}^{N \times d} \quad (14)$$

The composite training loss is:

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_{\text{link}} + \lambda \mathcal{L}_{\text{temporal}}, \quad \lambda = 0.1 \\ \mathcal{L}_{\text{link}} &= - \sum_{t=1}^T \sum_{(u,v)} [y_{uv} \log \sigma(\mathbf{e}_{u,t}^\top \mathbf{e}_{v,t}) + (1 - y_{uv}) \log(1 - \sigma(\mathbf{e}_{u,t}^\top \mathbf{e}_{v,t}))] \\ \mathcal{L}_{\text{temporal}} &= \sum_{t=2}^T \|\mathbf{E}_t - \bar{\mathbf{E}}_{<t}\|_F^2, \quad \bar{\mathbf{E}}_{<t} = \sum_{s<t} w_s \mathbf{E}_s, \quad w_s \propto e^{-0.5(t-1-s)} \end{aligned} \quad (15)$$

where $y_{uv} \in \{0, 1\}$ is the edge label and $\bar{\mathbf{E}}_{<t}$ is the recency-weighted EMA of all past spell embeddings (detached from the computation graph). Training uses AdamW (lr 10^{-3} , weight decay 10^{-4}) with 10% linear warmup followed by cosine annealing to 10^{-5} ; gradients are clipped to unit norm.

Shift prediction is post-hoc (no learned prediction head). Observed cosine shifts $c_{v,t} = 1 - \cos(\mathbf{e}_{v,t-1}, \mathbf{e}_{v,t})$ are computed over the training window for each term v , and the predicted next-spell

shift is the recency-weighted EMA with decay $\gamma = 0.7$:

$$\hat{y}_v = \sum_{t=2}^T w_t c_{v,t}, \quad w_t \propto \gamma^{T-t}, \quad \hat{y}_v = \max(0, \hat{y}_v) \quad (16)$$

APPENDIX C: FULL BENCHMARK LEADERBOARD

Table C1 reports all 38 benchmark models with complete evaluation metrics, grouped by experiment tier. NP denotes null predictor (near-zero MAE achieved by predicting zero shift for all terms). All delta values compare to DGT MAE = 0.077.

Table 11: Table C1. Full Benchmark Leaderboard (38 Models)					
Model	MAE	RMSE	Top-50	Δ	Sig.
<i>Experiment 1: Non-Neural Classic</i>					
GloVe Log-Cooccurrence SVD	0.121	0.204	0.00	+0.044	Yes
PPMI + SVD + Procrustes	0.132	0.212	0.06	+0.055	Yes
SGNS Shifted PPMI SVD	0.132	0.211	0.02	+0.055	Yes
Kleinberg Burst Approx.	1.313	1.738	0.10	+1.236	Yes
ARIMA Trend (NP)	0.000	0.000	0.04	−0.077	†
Degree Trend (NP)	0.000	0.000	0.02	−0.077	†
ETS Trend (NP)	0.000	0.000	0.04	−0.077	†
TF-IDF Slope (NP)	0.000	0.000	0.06	−0.077	†
PageRank Trend (NP)	0.000	0.000	0.04	−0.077	†
Raw Frequency Slope (NP)	0.000	0.000	0.04	−0.077	†
Weighted Degree Trend (NP)	0.000	0.000	0.02	−0.077	†
<i>Experiment 2: Graph Embedding and Dynamic GNN</i>					
DeepWalk SVD	0.111	0.207	0.00	+0.033	Yes
Node2Vec Biased Walk	0.121	0.203	0.02	+0.044	Yes
LINE First Order	0.149	0.244	0.00	+0.072	Yes
LINE Second Order	0.150	0.271	0.02	+0.073	Yes
GAT Link Reconstr. (NP)	0.031	0.108	0.00	−0.046	†
GCN Link Reconstr. (NP)	0.032	0.117	0.04	−0.045	†
GraphSAGE Link Reconstr. (NP)	0.030	0.103	0.06	−0.047	†
CAWN Causal Walk	0.102	0.167	0.04	+0.025	Yes
DySAT Structural Temporal	0.111	0.175	0.04	+0.034	Yes
EvolveGCN Temporal Smooth	0.086	0.154	0.02	+0.009	Yes
GraphMixer Event Mix	0.115	0.179	0.02	+0.037	Yes
JODIE Projected Drift	0.061	0.135	0.00	−0.016	No
TGAT Time-Weighted Edges	0.119	0.188	0.02	+0.041	Yes
TGN Memory Smooth	0.074	0.146	0.02	−0.003	No
<i>Experiment 3: Modern Graph Transformers</i>					
GraphGPS LapPE Proxy (NP)	0.030	0.103	0.06	−0.047	†
TokenGT Graph Token SVD	0.138	0.250	0.02	+0.061	Yes
<i>Experiment 4: Ablations</i>					
No Cumulative ETG	0.149	0.244	0.00	+0.072	Yes
k -Precedence Window ($k = 4$)	0.149	0.244	0.00	+0.072	Yes
Same-Post Co-occurrence	0.149	0.244	0.00	+0.072	Yes
Direct Precedence Window	0.124	0.206	0.00	+0.047	Yes
Unweighted Edges	0.124	0.206	0.00	+0.047	Yes
No Masked Attention (NP)	0.000	0.000	0.00	−0.077	†
No Laplacian PE (NP)	0.053	0.120	0.26	−0.025	†
No Temporal Attention	0.132	0.212	0.06	+0.055	Yes
Static GT Only (NP)	0.036	0.059	0.26	−0.041	†
DGT Full (proposed)	0.077	0.178	1.00	—	—
<i>Note: MAE and RMSE rounded to 3 decimal places. NP = null predictor (Top-50 overlap ≤ 0.06 and MAE < 0.03). Sig. = DGT significantly better at FDR $q < 0.05$ († = null predictor, not applicable). All models trained/evaluated on the same 6,000-term vocabulary and same held-out Spell 12 shift target.</i>					